

Thong Nguyen

✉ thongng96@gmail.com — ☎ +47 46364337 — 📍 Oslo, Norway —  —  —  — 

EXPERIENCE

- **Zalopay (VNG Corp.)**  — **Supervisor: Kien Tran**  Ho Chi Minh, Vietnam
April 2025 - Feb 2026
 - **Credit Scoring Modelling:** Developed a LightGBM credit scorecard model for Cashloan, achieving monthly Gini > 0.35 on 100K+ users
 - **Feature Engineering:** Engineered 3,000+ features from transactional, demographic, device, geolocation, and credit bureau data; optimized to $\sim 1,000$ key features
 - **Creating Dashboard on Tableau:** dashboards to monitor ML performance metrics (Gini, KS, PSI, drift) across 5+ partners/products
 - **Insight Analysis:** Conducted model iteration, ablation studies, and feature stability analysis to improve generalization across monthly cohorts
 - Collaborated with Analytics, Engineering, Product, and Risk teams to deploy monitoring pipelines and business insights
- **Tordivel AS**  — **Supervisor: Nell Trico**  Oslo, Norway
April 2022 - June 2025
 - **Data and Annotation Verification:** Led end-to-end CV data pipelines including collection, annotation, and preprocessing for industrial inspection tasks
 - **Modelling:** Developed and deployed deep learning models (Mask R-CNN, EfficientNet, U-Net) with mIoU of 0.85–0.9 across diverse datasets. Tuned key hyperparameters such as Anchor Boxes for dataset-specific optimization in object detection tasks
 - **Deployment:** Worked with engineering teams to integrate models into production software and standardize evaluation/monitoring pipelines
- **University of Eastern**  — **Supervisor: Joni Hyttinen**  Joensuu, Finland
Jan 2022 - Aug 2022
 - **Topic:** Visualization Solutions for Intraoperative Hyperspectral Imaging - [\[Link\]](#)
 - Proposed the DABS (Dual-Attention Band Selection) framework for hyperspectral image dimensionality reduction and trained **U-Net/HRNet** models for human placenta tissue segmentation.
 - Reduced hyperspectral bands by 45.9% ($37 \rightarrow 20$) while preserving spectral representation.
 - Developed image-based and patch-based training strategies, achieving 0.8 mIoU on the test set.
 - Built a **PyQt5** desktop application for segmentation visualization.
- **Chromasens GmbH (now: Allied Vision Konstanz GmbH)**  — **Supervisor: Timo Eckhard**  Konstanz, Germany
Jun 2021 - Aug 2021
 - Reviewed 30+ papers on illumination and inspection algorithms for industrial imaging.
 - Developed Halcon-based methods to detect dust, wafer cracks, and busbar defects in solar cells.
- **Emage Development Co. Ltd**  — **Tinh Van**  Ho Chi Minh City, Vietnam
Jan 2019 - Jul 2020
 - Developed deep learning solutions (**Mask R-CNN**, **EfficientNet**) for defect inspection in semiconductor manufacturing.
 - Worked with industrial imaging data including electronic chips, camera lenses, and microelectronic circuits.
 - **Built annotation, training, and validation software** using **WPF/C#** and integrated DL models with C++.
 - Prepared technical documentation and deployment support.

PUBLICATION

Milan Kresovic, **Thong Nguyen**, Mohib Ullah, Hina Afridi, Faouzi Alaya Cheikh. “PigPose: A Realtime Framework for Farm Animal Pose Estimation and Tracking” (2022) . IFIP International Conference on Artificial Intelligence Applications and Innovations.

SKILLS

- **Languages:** Python, git, PySpark, Langchain, C++, C#, WPF, Halcon, R, MATLAB.
- **Tools:** OpenCV, Numpy, Tensorflow, Keras, Pytorch, Scikit-Learn, Matplotlib, PyQt5, LaTeX, Pandas, Visual Studio Code, RAG.
- **Specialisation:** Computer Vision, AI & Deep Learning, Machine Learning, LLM, Data Science, Data Analysis, Fintech Colour Science, Software Development
- **Spoken languages:** Vietnamese (native), English (fluent), Norwegian (Basic)

PROJECTS

- **Home Credit Scoring -  (2025):** Developed a LightGBM credit scoring pipeline with feature selection (Correlation, LASSO, Random Forest), achieving $+0.37$ Gini gain on the Home Credit dataset.
- **HSI Placenta Tissue Segmentation -  (2022):** Developed HRNet-based hyperspectral image segmentation for human placenta tissue, achieving 89% mIoU and building a PyQt5 visualization tool.
- **Nighttime Image Dehazing -  (2021):** Implemented GASDA and CycleGAN-based domain adaptation for nighttime image dehazing using real and synthetic hazy datasets.

EDUCATION

- **MSc. Computational Colour and Spectral Imaging (COSI)** August 2020 - September 2022
NTNU(1), UEF(2), UJM(3) - Erasmus Mundus Program  Norway, France, Finland
 - **Thesis:** Visualization Solutions for Intraoperative Hyperspectral Imaging.
 - (1) MSc in Computational Colour and Spectral Imaging – Norwegian University of Science and Technology.
 - (2) MSc in Computer Science – University of Eastern Finland.
 - (3) MSc in Optics, Image, Vision, Multimedia – Jean Monnet University.
 - GPA: 3.44 (Good)
 - Courses: Advanced Image Processing, Deep Learning, Machine Learning, Statistics Data Mining, 3D Models in Computer Vision, Spectral Imaging.
- **BEng. Automation and Control Engineering Technology** August 2014 - April 2019
Ho Chi Minh city University of Technology and Education - GPA: 7.6/10 Vietnam
- **BSc. Electrical Engineering** September 2017 - December 2017
University College Cork - One semester exchange. Fully funded by SHARE Scholarship.  Ireland

HONORS AND AWARDS

- Student representative of the Master program - COSI (2020-2022).
- Erasmus Mundus Master Joint Degree - COSI program fee-waiver & Erasmus Mobility Scholarship (2020)
- Selected speaker for Boosting Education in ASEAN workshop, in Phnom Penh, Cambodia (2018).
- Selected as one of 50 ASEAN students for one exchange semester in Europe fully funded by SHARE Scholarship (2017).